

TG ECET (WP)-2026**SYLLABUS FOR CIVIL ENGINEERING****MATHEMATICS (10 Marks)**

S.No	Syllabus
1	UNIT I: MATRICES AND DETERMINANTS Definition of Matrix-order of a matrix-Type of matrices - Algebra of matrices- sum, difference, scalar multiplication and product of 2x2 Matrix - Equality of two matrices- Transpose of a Matrix – Determinant of a square Matrix of order 2×2 .
2	UNIT II: CO-ORDINATE GEOMETRY Straight lines: Slope of straight line - Various forms of straight line: Point-Slope form, Two- Point form, Intercept form, and General equation of a Straight-line Conditions for Parallelism and Perpendicularity of the Straight lines – Equation of a straight line parallel and perpendicular to the given straight line -Distance between two Parallel lines.
3	UNIT III: TRIGONOMETRY Trigonometric Ratios of Compound angles (without proof) - Trigonometric Ratios of Multiple and Sub- multiple angles ($2A$, $3A$, $A/2$).

CIVIL ENGINEERING (90 Marks)

S.No	Syllabus
1	UNIT-I: ENGINEERING MECHANICS Forces and moments- Vectors and scalars (Without i,j,k representations), resultant forces at a point, resolution and moment of forces, types of supports, beams, types of loading. Centroid – rectangle, triangle, parallelogram, circle, semicircle, trapezium. Location of centroid of T, L, I, channel, Z sections. built-up sections. Moment of Inertia and radius of gyration – T, L, I and built up sections,; Polar moment of inertia of solid and hollow circular sections using perpendicular axis theorem only. Simple stresses and strains - ductile materials- - Hooke's law - lateral strain-Poisson's ratio-Elastic constants and the relation between them- Composite sections-Mechanical properties of materials.
2	UNIT-II: STRENGTH OF MATERIALS Shear force and Bending Moment Diagrams for cantilever, Simply supported and overhanging beams subjected to Point loads and UDL. Theory of simple bending-assumptions-bending equation-bending Stresses-Section Modulus Neutral Axis - Flexural Rigidity, Radius of curvature, Moment of Resistance – Calculation of bending stresses in Symmetrical and Unsymmetrical sections-practical applications.Shear stress distribution across various sections like rectangular, circular and I and T- Sections-Torsion-solid and hollow circular shafts subjected to pure torsion – shear stress – distribution in shafts – power transmitted by circular shafts.

	Slope and Deflection of cantilevers and simply supported beams by Double Integration method and Deflection of simply supported beams by Macaulay's method – Mohr's theorems for slopes and deflections-Symmetrical loading. Columns and struts-types-slenderness ratio- Euler's and Rankine's formulae for axial loading.
3	UNIT-III: REINFORCED CONCRETE STRUCTURES Grades of concrete, characteristic strength, Modulus of Elasticity-I.S.456-2000- Philosophy of Limit state design. Limit state of Strength and Serviceability, partial safety factor-design strength of materials and design loads- assumptions – stress strain diagram of singly reinforced R.C.C beams. Analysis and Limit state design of rectangular Beams-Singly, Doubly reinforced and T-beams. Shear in RCC beams - Development length. Slabs-analysis and limit state design of one-way and two-way slabs as per IS.456-2000, Torsion reinforcement. Design of continuous slabs and beams - Deflection check for slabs and beams. Detailing of reinforcement in singly reinforced and doubly reinforced simply supported beams of rectangular sections and lintels, one way and two way slabs. Columns: Codal provisions of I.S 456-2000 - short and long columns-different shapes-design of short columns by limit state method-long columns- concept, effective length for different end conditions. Footings-Isolated column footings-one way shear and two way shear.
4	UNIT-IV: SURVEYING Chain surveying - purpose and principle-offsets - errors and corrections- different operations in chain surveying- obstacles - methods of calculation of area. Compass Surveying - purpose and principle - bearings- traversing using prismatic compass- local attraction - errors. Levelling - definitions - component parts of Dumpy level - contouring - Methods of levelling - errors - characteristics and methods. Theodolite - principles and component parts- fundamental lines and relationship among them – temporary adjustments of theodolite - measurement of horizontal and vertical angles - errors-traverse computations - Bowditch and Transit rule. Tachometry - principles - stadia tachometry - tangential tachometry(simple problems), Principle and uses of E.D.M, Electronic Theodolite, Total Station, Global positioning System - Importance, G.I.S – Use and applications in Civil Engineering, Curves - simple curves, elements of simple, compound and reverse curves, setting out of simple curves by chain & tape, single & double theodolite method(problems).
5	UNIT-V: HYDRAULICS Fluid properties - specific weight – mass density-specific gravity - surface tension - capillarity-viscosity. Atmospheric pressure, gauge pressure and absolute pressure. Centre of pressure and total pressure on plane surfaces, measurement of fluid pressure using piezometer and manometers. Types of flows-uniform, non-uniform, steady, unsteady, laminar and turbulent flows. Energies of liquid in motion - continuity equation. Bernoulli's theorem - Pitot tube - Venturimeter. Flow through small and large orifices, coefficients of orifices - C_c, C_v and C_d. Flow through internal, external, convergent and divergent mouthpieces. Types of Notches – rectangular, triangular and trapezoidal notch, Types of Weirs- sharp crested and broad crested-mathematical formulae for discharge- Francis and

	Bazin's empirical formulae. Flow through pipes-major and minor losses - Chezy's and Darcy's formulae for loss of head due to friction-HGL & TEL. Discharge through parallel pipes, Reynolds number, Flow through open channels-rectangular and trapezoidal - Chezy's formula for discharge - Manning's equation for Chezy's constants-Most economical sections. Reciprocating and Centrifugal pumps (without problems). Classification of Turbines - Kaplan, Francis and Pelton wheel (without problems) – component parts and functions of surge tanks.
6	UNIT VI: IRRIGATION ENGINEERING Necessity of Irrigations - Perennial and inundation Irrigation, Flow and Lift Irrigation, Principal seasons - kharif and rabi crops - Duty, delta and base period. Methods of Irrigation - check flooding, basin flooding, Border strip, furrow, sprinkler and drip Irrigations. Hydrology - Rainfall, types of Rain gauges, types of catchments-rainfall and runoff (without problems). Measurement of velocity of flow in streams. Classification of Head works - component parts of diversion head works. Weirs and Barrages. Percolation and uplift pressures. Types of Reservoirs - dead storage and live storage. Storage Head works-different types of dams- gravity dams-low and high dams. Failures of gravity dams - drainage galleries. Types of spillways. Earth dams - types, failures and precautions. Phreatic lines. Distribution works-classifications and alignment of canals-typical cross section of canal-berms - balanced depth of cutting- canal lining. Cross drainage works – types and functions. Soil erosion, Types and causes-measures to control erosion.
7	UNIT VII: TRANSPORTATION ENGINEERING Importance of transportation engineering – Highway development in India, I.R.C. – Classification of roads as per I.R.C., alignment of Highway & Highway surveys, recommended I.R.C. values of camber for different roads. Gradients – Ruling gradient, limiting and exceptional gradient, Recommended I.R.C values of gradients. Super elevation & sight distance. Traffic Engg. - Traffic studies and its importance, Road intersections- Traffic signs-Informatory signs- Mandatory signs, Cautionary signs – pavement – types & its design. Highway constructions and Maintenance. Tests on road materials - Purpose of road drainage - surface and sub-surface drainage, Typical cross section of highway in cutting and embankment. Water bound macadam roads, Cement concrete roads. Permanent way of Railways, Importance of Railways- Gauge, Types of gauges, Structure of permanent way – different types of rails, requirements of a good rail, Sleepers - functions, Types of sleepers, characteristics of a good sleeper – spacing of sleepers-sleeper density (without problems). Rail joints, coning of wheels, ballast - Points and crossings, turnout.
8	UNIT VIII : WATER SUPPLY AND SANITARY ENGINEERING Quality of water, Need for protected water supply, Total quantity of water for a town, per capita demand and factors affecting demand, Forecasting population by arithmetical, geometrical and incremental increase methods, Sources and conveyance of water: surface sources, underground sources, Types of Intakes. Quality and Methods of purification of water. Distribution System: Methods of supply, Storage-Distribution systems, Types of layout-

	dead end, grid, radial and ring system their merits and demerits and their suitability. General layout of water supply arrangements in buildings. System of sewage disposal-types of sewerage systems, design of sewers. Different shapes of cross-section for sewers, Strength of sewage, sampling of sewage, characteristics of sewage - principles of treatment, Preliminary and secondary treatment (theoretical concepts). BOD-COD - pH. Sewers – sewer appurtenances - shapes, merits and demerits. Septic tank – soak pit (without problems).
9	UNIT IX: BUILDING MATERIALS AND CONSTRUCTION PRACTICE Stones-classification of rocks. Bricks – manufacturing, tests on bricks. Tiles- types of tiles. Cement- classification, manufacturing - tests - As per IS specifications. Mortars – classification - proportioning. Concrete- proportioning – water-cement ratio – workability – admixtures-curing methods-R.M.C. Timber and surface protective materials. Characteristics-types and uses. Classification of buildings, foundations-N.B.C classification-bearing capacity of soil- types of foundations. Masonry-Bonds in brick masonry. Plastering-purpose. Pointing purpose and types.